As before, if x_{pre} (0.85 mm lead) is subtracted from this value, the same result (1.4 mm lead) is obtained.

Now, consider a wall adjacent to the wall on which the chest image receptor in the chest room is mounted. This wall is never struck by the primary beam and is therefore a secondary barrier. Assume a fully occupied, uncontrolled (P/T = 0.02 mGy week^{-1}) area located a distance d_{sec} = 2.1 m from both the patient and x-ray tube. From Table 4.7, the total unshielded secondary air kerma for leakage plus 90 degree scatter (side-scatter) at 1 m from the chest unit is 2.7 × 10^{-3} mGy patient^{-1}. Then, from Equation 4.4, the weekly unshielded secondary air kerma would be:

$$K_{sec}(0) = \frac{2.7\times10^{-3}\ \text{mGy patient}^{-1}\times 300\ \text{patients week}^{-1}}{(2.1\ \text{m})^2} = 0.18\ \text{mGy week}^{-1}.$$

The secondary barrier transmission is therefore:

$$B_{sec}(x_{barrier}) = \frac{0.02\ \text{mGy week}^{-1}}{0.18\ \text{mGy week}^{-1}} = 1.1\times10^{-1}.$$

From Figure C.2, this transmission would be obtained for a dedicated chest unit by a 0.42 mm thick lead barrier. As before, the calculation can also be made using Equation 4.10 and the secondary transmission fitting parameters (α, β and γ) from Table C.1:

$$x_{barrier} = \frac{1}{2.288\times1.054}\ln\left[\frac{\left(\frac{300\times1\times0.0027}{0.02\times2.1^2}\right)^{1.054}+\frac{9.848}{2.288}}{1+\frac{9.848}{2.288}}\right] = 0.42\ \text{mm}.$$

The nearest commercially available lead sheet ≥0.42 mm sheet is 1/32 inch (0.79 mm) (Figure 2.3). Again the adequacy of both the primary and secondary barriers in achieving the effective dose limit for members of the public is confirmed by means of the performance assessment by the qualified expert.

5.3 The Radiographic Room

Consider next the radiographic room in Figure 5.1 (elevation drawing) and Figure 5.2 (plan drawing). Assume N = 125 patients per week are radiographed in this room. The workload distribution is assumed to follow that of the radiographic room from the AAPM-TG9 survey (Simpkin, 1996a).The areas exposed to primary radiation include the office beneath the floor, the staff rest room adjacent to the chest image receptor, and the cross-table wall

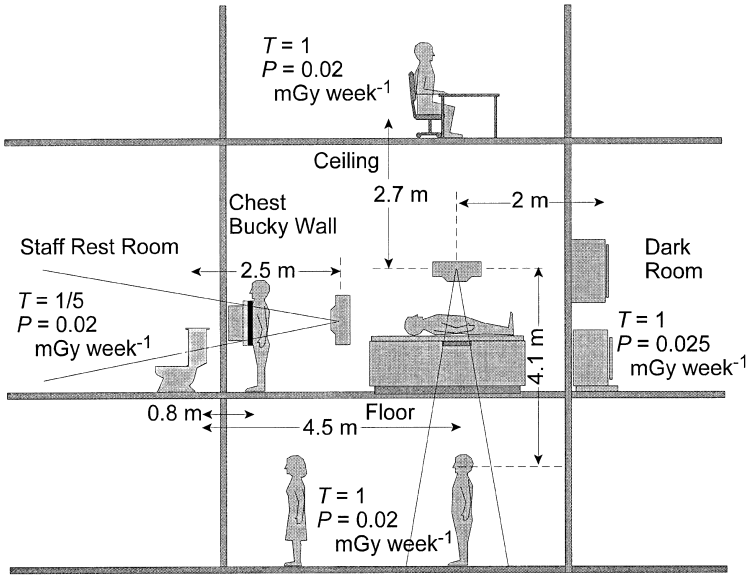


Fig. 5.1. Elevation drawing of the radiographic room. The dimensions are used in sample calculations in Section 5.3. This same layout is also used for the R&F room examples in Section 5.4, with the addition of a fluoroscopy x-ray tube beneath the table and an image intensifier over the table.

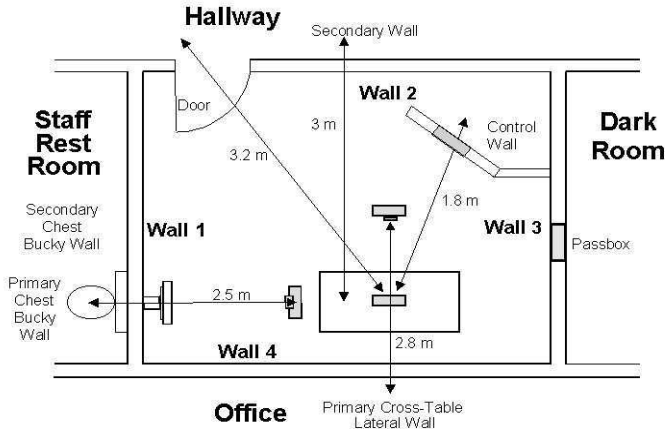


Fig. 5.2. Plan drawing of the radiographic room shown in Figure 5.1.

(Wall 4 behind the x-ray table as shown in Figure 5.2). All other areas are assumed to be exposed only to secondary radiation. The values for other shielding parameters that need to be determined by the qualified expert for this room are noted either in Figure 5.1, Figure 5.2, or in the examples.

5.3.1 *The Floor of the Radiographic Room*

Assume the area below the radiographic room is an uncontrolled area with a shielding design goal of $P = 0.02$ mGy week^{-1} with an occupancy factor $T = 1$. This area will be irradiated by primary radiation directed at the image receptor in the radiographic table, as well as by secondary radiation.

5.3.1.1 *Primary Barrier Calculation for Floor Beneath the Radiographic Table.* From Table 4.5, the unshielded primary air kerma per patient for the *Rad Room (floor or other barriers)* workload distribution is 5.2 mGy patient^{-1} at 1 m. While the use factor (U) for this workload distribution directed on the floor is 0.89 (Table 4.4), one may use a conservatively safe assumption that $U = 1$. Thus, at the location 1.7 m above the lower floor (Section 4.1.2) (*i.e.*, $d_P = 4.1$ m), the total unshielded primary air kerma per week, from Equation 4.3, is:

$$K_P(0) = \frac{5.2 \text{ mGy patient}^{-1} \times 1 \times 125 \text{ patients week}^{-1}}{(4.1 \text{ m})^2} = 39 \text{ mGy week}^{-1}.$$

The primary barrier transmission required for $T = 1$ and $P = 0.02$ mGy week^{-1} is:

$$B_P(x_{barrier} + x_{pre}) = \frac{0.02 \text{ mGy week}^{-1}}{39 \text{ mGy week}^{-1}} = 5.1 \times 10^{-4}.$$

Use of the primary transmission curve for concrete (Figure B.3) for the *Rad Room (floor or other barriers)* workload distribution, results in a required total thickness ($x_{barrier} + x_{pre}$) of 107 mm concrete. From Table 4.6, the attenuation provided by a typical radiographic table and image receptor (ignoring patient attenuation) is equivalent to 72 mm concrete. Thus, $x_{pre} = 72$ mm concrete and the net thickness required in the floor under the x-ray table to attenuate the primary beam is $x_{barrier} = (107 - 72 \text{ mm}) = 35$ mm. These results may also be obtained from Equation 4.8 using the fitting parameters in Table B.1 for the primary beam transmission for the *Rad Room (floor or other barriers)* workload distribution.

This result may also be quickly arrived at using the method described in Section 4.2.4. Using the same parameters as before, the first step is to determine the value of NT/Pd^2:

$$\frac{NT}{Pd^2} = \frac{125 \text{ patients week}^{-1} \times 1}{0.02 \text{ mGy week}^{-1} \times (4.1 \text{ m})^2} = 372 \text{ mGy}^{-1} \text{ m}^{-2}.$$

Then, using this value and Figure 4.6b (to account for the attenuation from the image receptor and radiographic table), a net barrier thickness of 37 mm concrete is required. This calculation includes the secondary radiation present from procedures done against the chest bucky and cross-table lateral work. This result is similar to the 35 mm concrete barrier calculated previously. However, the first calculation only considered the primary beam contribution from the over-table tube.

5.3.1.2 *Secondary Barrier Calculation for Floor.* Floor areas away from the table need to serve as a secondary barrier for exposures directed at the patient on the table and chest image receptor and their shielding adequacy needs to be verified. For example, shielding required at the location of the woman in Figure 5.1 needs to be determined. There are two independent secondary radiation sources that need to be considered, namely the patient on the radiographic table and the patient against the chest image receptor. Note that this secondary radiation will be assumed to impact the floor directly without attenuation by the table-mounted image-receptor hardware. The workload distribution *Rad Room (all barriers)* that includes scattered and leakage radiations from both sources is utilized. A conservatively safe assumption is that the x-ray tube is located so that the scattered and leakage radiation distances are equal to the vertical distance from the patient to the location of the woman in Figure 5.1 (*i.e.*, $d_S \approx d_L \approx d_{sec} = 3$ m) (3 m not indicated on Figure 5.1). From Table 4.7, the total unshielded secondary air kerma per patient at 1 m, using the *Rad Room (all barriers)* workload distribution for leakage plus 90 degree scatter (side-scatter), is 3.4×10^{-2} mGy. Thus, from Equation 4.4, the unshielded secondary air kerma for 125 patients per week is:

$$K_{sec}(0) = \frac{3.4 \times 10^{-2} \text{ mGy patient}^{-1} \times 125 \text{ patients week}^{-1}}{(3 \text{ m})^2} = 0.47 \text{ mGy week}^{-1}.$$

To reduce this to 0.02 mGy week^{-1}, a secondary barrier transmission of:

$$B_{sec}(x_{barrier}) = \frac{0.02 \text{ mGy week}^{-1}}{0.47 \text{ mGy week}^{-1}} = 4.3 \times 10^{-2}$$

is required. Using Figure C.3 for the transmission of secondary radiation through concrete and utilizing the curve for the *Rad Room (all barriers)* workload distribution, the concrete floor thickness required beyond the radiographic table is 33 mm. Hence, the 35 mm concrete thickness required from the primary barrier calculation under the radiographic table will suffice for the entire floor.

This result can be reproduced using the method described in Section 4.2.4. Here d = 3 m. As before, first determine the value of NT/Pd^2:

$$\frac{NT}{Pd^2} = 694 \text{ mGy}^{-1} \text{ m}^{-2}.$$

Then, using Figure 4.6c for the off-table secondary floor, it is seen that a barrier thickness of 33 mm concrete is again found.

5.3.2 *The Ceiling of a Radiographic Room*

This area is uncontrolled (P = 0.02 mGy week^{-1}) with an occupancy factor T = 1. This barrier is purely a secondary barrier. Assume, as above, that only one x-ray tube location is needed, with d_L = 2.7 m and d_S = 3.5 m (3.5 m not indicated on Figure 5.1). To be conservatively safe, set $d_S = d_L = d_{sec}$ = 2.7 m. Assuming leakage plus forward/backscatter (a conservatively high assumption) from Table 4.7, it is found that the total unshielded air kerma per patient is 4.9×10^{-2} mGy for the *Rad Room (all barriers)* workload distribution. The unshielded total air kerma, from Equation 4.4, is then:

$$K_{sec}(0) = \frac{4.9 \times 10^{-2} \text{ mGy patient}^{-1} \times 125 \text{ patients week}^{-1}}{(2.7 \text{ m})^2} = 0.84 \text{ mGy week}^{-1}.$$

To reduce this to 0.02 mGy week^{-1} requires a secondary barrier transmission of:

$$B_{sec}(x_{barrier}) = \frac{0.02 \text{ mGy week}^{-1}}{0.84 \text{ mGy week}^{-1}} = 2.4 \times 10^{-2}.$$

Figure C.3 in Appendix C then yields a required barrier thickness of 44 mm concrete for the ceiling. Note that the distance from the patient to the occupied area (d_S) was assumed to be the same distance as the distance from the x-ray tube head (d_L), the closer of the two sources of secondary radiation.

Alternatively, using the method of Section 4.2.4 and d = 2.7 m for this barrier, then:

$$\frac{NT}{Pd^2} = 857 \text{ mGy}^{-1} \text{ m}^{-2}.$$

From Figure 4.6c, the required ceiling thickness is found to be 39 mm of concrete. This result is slightly lower than that calculated previously, since Figure 4.6c was generated using more accurate distances from the patient at the chest bucky and the table to the ceiling. These slightly greater distances diminish the scattered radiation contribution to the air kerma at the ceiling, thereby allowing a thinner barrier.

5.3.3 *Wall Containing the Chest Image Receptor in the Radiographic Room*

As shown in Figure 5.1, the area behind the chest image receptor is a staff rest room. Since employees who do not work with radiation sources also use this rest room, the shielding design goal for an uncontrolled area applies, namely P = 0.02 mGy week^{-1}. From Table 4.1, the suggested occupancy factor for a staff rest room is T = 1/5. Therefore, P/T = 0.1 mGy week^{-1}.

5.3.3.1 *Primary Barrier: Chest Image Receptor.* The use factor for the *Rad Room (chest bucky)* workload distribution is U = 1 for exposures made on the chest image receptor. As explained in Section 4.1.4, N is also 125 patients week^{-1} for this barrier. Using Table 4.5 with the *Rad Room (chest bucky)* workload distribution, the weekly unshielded primary air kerma at 2.5 m from the chest tube position, from Equation 4.3, is:

$$K_P(0) = \frac{2.3 \text{ mGy patient}^{-1} \times 1 \times 125 \text{ patients week}^{-1}}{(2.5 \text{ m})^2} = 46 \text{ mGy week}^{-1}.$$

The primary barrier transmission is then:

$$B_P(x_{\text{barrier}} + x_{\text{pre}}) = \frac{0.1 \text{ mGy week}^{-1}}{46 \text{ mGy week}^{-1}} = 2.2 \times 10^{-3}.$$

From Figure B.2, the required total ($x_{\text{barrier}} + x_{\text{pre}}$) lead thickness for this workload distribution is 1.3 mm. From Table 4.6, the attenuation provided by a typical wall-mounted image receptor is equivalent to x_{pre} = 0.85 mm of lead. The recommended wall shielding therefore would be x_{barrier} = 0.45 mm lead, and 0.79 mm (1/32 inch) lead (the thinnest available thickness of sheet lead) should be specified.

One may also use the methodology of Section 4.2.4. Here d = 2.5 m from the chest x-ray tube to the occupied area. Then:

$$\frac{NT}{Pd^2} = 200 \text{ mGy}^{-1} \text{ m}^{-2}.$$

From Figure 4.5b, a barrier of 0.5 mm lead is indicated. This is in agreement with the calculation above.

5.3.3.2 *Secondary Barrier: Chest Image-Receptor Wall*. The area of the staff rest room that is outside the primary beam is irradiated by secondary radiation that is not attenuated by the chest image receptor. There are two scattered and leakage radiation sources to consider. One is the secondary radiation generated by the over-table exposures. The other is the secondary radiation from exposures made against the chest image receptor itself.

The unshielded secondary air kerma from the over-table x-ray tube location can be determined using Table 4.7. Assume leakage radiation plus side-scattered radiation, and the *Rad Room (floor or other barriers)* workload distribution with d_{sec} = 4.5 m. Then, from Equation 4.4:

$$K_{sec}(0) = \frac{2.3 \times 10^{-2} \text{ mGy patient}^{-1} \times 125 \text{ patients week}^{-1}}{(4.5 \text{ m})^2} = 0.14 \text{ mGy week}^{-1}.$$

The scattered and leakage radiations due to exposures made against the chest image receptor should be considered independently, since the scattered and leakage radiation distances are significantly different. Let the scattered radiation distance from the patient against the chest image receptor to the occupied area be d_S = 0.8 m. The leakage radiation distance from the x-ray tube to this area is d_L = 2.5 m. Then from Table 4.7, for the *Rad Room (chest bucky)* workload distribution, the unshielded side-scattered and leakage air kermas from these sources are 4.9×10^{-3} and 3.9×10^{-4} mGy patient^{-1}, respectively. From Equation 4.4, the sum of these contributions is:

$$K_{sec}(0) = \left[\frac{4.9 \times 10^{-3} \text{ mGy patient}^{-1}}{(0.8 \text{ m})^2} + \frac{3.9 \times 10^{-4} \text{ mGy patient}^{-1}}{(2.5 \text{ m})^2}\right] \times 125 \text{ patients week}^{-1}$$

or

$$K_{sec}(0) = 0.96 + 0.008 = 0.97 \text{ mGy week}^{-1}.$$

To this sum is added the previously calculated secondary radiation from the over-table tube location. Thus, the total unshielded secondary air kerma is:

$$K_{sec}(0) = 0.97 + 0.14 = 1.1 \text{ mGy week}^{-1}.$$

and the required barrier transmission factor is:

$$B(x_{barrier}) = \frac{0.1 \text{ mGy week}^{-1}}{1.1 \text{ mGy week}^{-1}} = 9.1 \times 10^{-2}.$$

The greatest contribution to the secondary air kerma is due to exposures against the chest bucky. Therefore, for simplicity and to be conservatively safe, assume the more penetrating *Rad Room (chest bucky)* workload distribution. From Figure C.2, a barrier of 0.35 mm lead is required. A more realistic calculation using the correct location for each scattered or leakage radiation source and a 30 degree scattering angle for the chest source with the correct workload distribution for each tube location yields 0.3 mm lead.

One may also use the shielding graphs from Section 4.2.4. Here d = 2.5 m from the chest tube to the occupied area. Substituting N = 125 patients week^{-1} and P/T = 0.1 mGy week^{-1} then:

$$\frac{NT}{Pd^2} = 200 \text{ mGy}^{-1} \text{ m}^{-2}.$$

From Figure 4.5c, a barrier of 0.37 mm lead is obtained, which is in good agreement with the values given above.

Since the primary shielding is greater than the secondary wall requirements, the entire wall can be shielded with a minimum of the primary requirement, 0.45 mm or 1/32 inch lead, the nearest greater standard thickness.

5.3.4 *Darkroom Wall in the Radiographic Room*

In shielding a darkroom, the limiting factor is usually the exposure to the film stored in the darkroom rather than to personnel occupying the darkroom. The recommended limit for unexposed film stored in boxes or in the film bin is an air kerma of 0.1 mGy during the storage period (Suleiman *et al.*, 1995). Assuming a one-month storage period, this is 0.025 mGy week^{-1}. Note that, since film storage shelving is often located higher than the usual shielding height of 2.1 m above the floor, it may be desirable to extend the shielding in this wall to at least 2.5 m above the floor.

For the radiographic room in Figure 5.1, assume that no exposures are made with the primary beam directed toward the darkroom wall, so that only secondary radiation need be considered. For simplicity, assume that all secondary radiation is generated with the x-ray tube over the table at a secondary distance of 2 m. The *Rad Room (all barriers)* workload distribution is appropriate for this case. Then, from Table 4.7 using the value for leakage radiation plus side-scattered radiation, the unshielded secondary air kerma in the darkroom is, from Equation 4.4:

$$K_{sec}(0) = \frac{3.4 \times 10^{-2}\ \text{mGy patient}^{-1} \times 125\ \text{patients week}^{-1}}{(2\ \text{m})^2} = 1.1\ \text{mGy week}^{-1}.$$

The required barrier will have transmission:

$$B_{sec}(x_{barrier}) = \frac{0.025\ \text{mGy week}^{-1}}{1.1\ \text{mGy week}^{-1}} = 2.3 \times 10^{-2}.$$

The required shielding from Figure C.2 is therefore 0.53 mm lead.

In like manner, the shielding curves from Section 4.2.4 (Figure 4.5c) may be used. Letting d = 2 m for this darkroom wall:

$$\frac{NT}{Pd^2} = 1{,}250\ \text{mGy}^{-1}\ \text{m}^{-2}.$$

From Figure 4.5c for a secondary wall, a barrier with a thickness of 0.53 mm lead is required as before.

The film passbox between the darkroom and the radiographic room will typically contain unexposed film loaded in cassettes. This greatly increases the sensitivity of the film to radiation-induced fogging by factors in excess of 100 (Suleiman *et al*., 1995). Assuming that all the cassettes in the passbox will be recycled in 1 d (*e.g.*, once every 25 patients), the unshielded air kerma over 1 d to a cassette in the passbox would then be:

$$K_{sec}(0) = \frac{3.4 \times 10^{-2}\ \text{mGy patient}^{-1} \times 25\ \text{patients}}{(2\ \text{m})^2} = 0.21\ \text{mGy}.$$

Assuming an air kerma of 0.5 μGy (5×10^{-4} mGy) will fog a film in a cassette (Suleiman *et al*., 1995), the shielding for these cassettes requires a transmission of not more than:

$$B_{sec}(x_{barrier}) = \frac{5 \times 10^{-4}\ \text{mGy}}{0.21\ \text{mGy}} = 2.3 \times 10^{-3}.$$

From Figure C.2, this requires a lead barrier thickness of 1.3 mm on all sides (except the darkroom side) of the film passbox.

5.3.5 *The Cross-Table Wall in the Radiographic Room*

Now consider the wall behind the x-ray table (Wall 4 in Figure 5.2) referred to as the cross-table wall. Radiographic exposures will be directed at this wall with only the patient, grid and cassette in the primary beam. However, only a small fraction of the exposures made on the 125 patients per week will be imaged using the cross-table alignment. From Table 4.4, the use factor U for such a wall is 0.09. This is applied to the *Rad Room (floor or other barriers)* workload distribution. Assume the wall protects a fully occupied, uncontrolled office (P/T = 0.02 mGy week^{-1}) at primary distance d_P = 2.8 m from the x-ray tube. From Equation 4.3, the unshielded air kerma is:

$$K_P(0) = \frac{5.2\ \text{mGy patient}^{-1} \times 0.09 \times 125\ \text{patients week}^{-1}}{(2.8\ \text{m})^2} = 7.5\ \text{mGy week}^{-1}.$$

The required transmission for this wall is therefore:

$$B_P(x_{barrier} + x_{pre}) = \frac{0.02\ \text{mGy week}^{-1}}{7.5\ \text{mGy week}^{-1}} = 2.7 \times 10^{-3}.$$

From Figure B.2, the total lead thickness required to achieve this transmission is 0.9 mm lead. From Table 4.6, the lead-equivalent thickness of the grid- and cassette-image receptor (x_{pre}) is 0.3 mm. If proper collimation is assumed, the structural barrier thickness is then $x_{barrier}$ = (0.9 – 0.3 mm) = 0.6 mm lead.

However, this is not the total barrier requirement. Secondary radiation from the vertically directed beam can also reach the same areas as the primary cross-table beam. This secondary radiation may not be negligible compared to the primary due to the low-primary beam use factor and the preshielding by the grid and cassette.

In this area of the table, we can ignore the contribution from the chest tube location due to distance and oblique barrier penetration; hence, the *Rad Room (floor or other barriers)* workload distribution will be utilized for the secondary calculation. Assuming a secondary distance of 2 m with leakage radiation plus side-scattered radiation (2 m not indicated in Figure 5.2), the unshielded secondary air kerma, using Table 4.7 and Equation 4.4, is:

$$K_{sec}(0) = \frac{2.3 \times 10^{-2}\ \text{mGy patient}^{-1} \times 125\ \text{patients week}^{-1}}{(2\ \text{m})^2} = 0.72\ \text{mGy week}^{-1}.$$

The total transmitted air kerma by a barrier of thickness $x_{barrier}$ is then:

$$K_{tot}(x_{barrier}) = [0.72\ \text{mGy week}^{-1} \times B_{sec}(x_{barrier})] + [7.5\ \text{mGy week}^{-1} \times B_P(x_{barrier} + x_{pre})],$$

where $B_{sec}(x_{barrier})$ is the secondary and $B_P(x_{barrier} + x_{pre})$ the primary transmission. A closed solution for $x_{barrier}$ is not possible, even assuming the same transmission data applies to both primary and secondary source components due to x_{pre}. Trial solutions may be made by iteration to find the final thickness $x_{barrier}$ required to reduce K to $P/T = 0.02$ mGy week^{-1}.

An example of the procedure would be as follows for the case of $P/T = 0.02$ mGy week^{-1}. Since the primary beam alone with preshielding required 0.6 mm lead, the qualified expert would probably recommend 0.79 mm lead (standard 1/32 inch thickness). One approach is to determine if 0.8 mm lead is in fact adequate. Using $x_{barrier} = 0.8$ mm lead and $x_{pre} = 0.3$ mm lead, the total transmitted air kerma is:

$$K_{tot}(0.8\ \text{mm}) = (0.72\ \text{mGy week}^{-1} \times 5 \times 10^{-3}) + (7.5\ \text{mGy week}^{-1} \times 1.4 \times 10^{-3}) = 0.014\ \text{mGy week}^{-1}.$$

where transmission data has been obtained using Figure C.2 and Figure B.2 for the *Rad Room (floor or other barriers)* distribution. Therefore, 0.8 mm lead is adequate, resulting in <0.02 mGy week^{-1}. A more complete calculation utilizing all sources (but ignoring oblique incidence from the chest tube location) yields $x_{barrier} = 0.78$ mm lead.

The methodology of Section 4.2.4 also may be used to determine this barrier thickness. Using the distance $d = 2.8$ m from the cross-table lateral primary tube location to the occupied area:

$$\frac{NT}{Pd^2} = 797\ \text{mGy}^{-1}\ \text{m}^{-2}.$$

Accounting for image-receptor shielding from Figure 4.5b, a lead barrier 0.83 mm thick is required. If the image-receptor shielding is ignored, a lead barrier of 1.03 mm (Figure 4.5a) would be required.

5.3.6 *Control Wall in the Radiographic Room*

The area behind the control wall shown in Figure 5.2 is a controlled area and, as discussed in Section 2.2.1, this control booth region *should* be considered a secondary barrier. Assuming a secondary distance of 1.8 m and the leakage plus 90 degree side-scatter value for the *Rad Room (all barriers)* workload distribution from Table 4.7, the unshielded secondary air kerma for 125 patients per week, from Equation 4.4, is:

$$K_{sec}(0) = \frac{3.4 \times 10^{-2}\ \text{mGy patient}^{-1} \times 125\ \text{patients week}^{-1}}{(1.8\ \text{m})^2} = 1.3\ \text{mGy week}^{-1}.$$

To reduce this to the shielding design goal for a controlled area (P = 0.1 mGy week^{-1}), a secondary barrier transmission of:

$$B_{sec}(x_{barrier}) = \frac{0.1\ \text{mGy week}^{-1}}{1.3\ \text{mGy week}^{-1}} = 7.7 \times 10^{-2}.$$

is required. Use of Figure C.2 for the *Rad Room (all barriers)* workload distribution indicates that the required lead barrier thickness for the control booth is approximately 0.27 mm. This can be verified by using the methodology of Section 4.2.4. Here, NT/Pd^2 = 386 also requires a lead barrier thickness of 0.27 mm from the "Secondary Wall" curve in Figure 4.5c. The qualified expert can specify a conservatively safe thickness of 1/32 inch sheet lead for this control wall and a similar equivalent lead thickness of lead glass for the view window in this wall.

There is often an additional factor that will influence the design of a control booth to a greater extent than protection of the operator. As discussed in Section 5.3.4, Suleiman *et al*. (1995) have shown that fogging of x-ray film in a cassette will occur if it is exposed to 0.5 μGy (5×10^{-4} mGy) or more. Many facilities typically store loaded cassettes behind the control barrier in radiographic and R&F rooms. Assuming a recycling time of 1 d, during which time an average of 25 patients will be radiographed (1/5 the weekly workload), the control wall shielding required is calculated as follows:

$$K_{sec}(0) = \frac{3.4 \times 10^{-2}\ \text{mGy patient}^{-1} \times 25\ \text{patients}}{(1.8\ \text{m})^2} = 0.26\ \text{mGy}.$$

To reduce this to 5×10^{-4} mGy requires a secondary barrier transmission of:

$$B_{sec}(x_{barrier}) = \frac{5 \times 10^{-4} \text{ mGy}}{0.26 \text{ mGy}} = 1.9 \times 10^{-3}.$$

Use of Figure C.2 for the *Rad Room (all barriers)* workload distribution indicates that the required lead barrier thickness for the control booth is 1.3 mm. This cannot be verified by using the methodology of Section 4.2.4 since the NT/Pd^2 value (15,432) (here P/T is 5×10^{-4} mGy for 1 d) is well outside the range of values in the graph in Figure 4.5c.

Unless specific information indicating that loaded cassettes will not be stored behind the control booth, the 0.5 μGy limitation for loaded cassettes per storage period *should* be assumed. As discussed by Sutton and Williams (2000), shielding of computed radiography cassettes behind the control wall should not introduce any more restrictive limitations than film/screen storage because fewer cassettes are typically used and therefore use of these is more frequent.

5.4 Radiographic and Fluoroscopic Room

Use of a radiographic and fluoroscopic (R&F) room may vary widely from one facility to another. A larger facility may have a room dedicated to patients requiring fluoroscopy, whereas a smaller facility may use the room for fluoroscopic cases as well as for cases requiring only radiographic exposures. The latter is probably more common than the former. That is, after several fluoroscopic cases in the morning, the room may revert to a purely radiographic room for the remainder of the day. Most fluoroscopic cases will also involve the overhead radiographic tube for taking additional films covering a larger field-of-view than available with spot-film images or for cross-table lateral films.

A R&F room can therefore represent a complex set of workloads and workload distributions. A typical room used only for fluoroscopic cases would be used for about 20 patients per week (the average room usage from the clinical survey) with the workload in mA min week^{-1} being dominated by the fluoroscopy tube (about 13 mA min patient^{-1} for the fluoroscopy tube and 1.5 mA min patient^{-1} for the overhead tube). The secondary transmission curves for these two workload distributions indicate that the x-ray spectra are roughly equivalent and more penetrating than the x-ray spectra for the workload distribution for a purely radiographic room. This is reasonable since the use of contrast media will often require higher operating potentials.

In the following example, an under-table fluoroscopy tube will be assumed. The primary beam from the fluoroscopic tube is totally intercepted by the over-table image intensifier assembly